

Product Requirements Document

Antenna Recommender (ANT)

Document ID ANT-PLAN-PRD	Phase Planning	Version 1.0
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1. Product Overview

The Antenna Recommender is a responsive website that recommends the best antenna according to a user's technical requirements. Users enter parameters such as frequency, application type, and required gain; the system returns a primary recommendation, at least one alternative, and a side-by-side comparison of key specifications.

2. Target Users

- RF Engineers
- Communication Engineering students
- Researchers

3. Core Features

Feature	Description
Technical parameter input	Users input required technical values, e.g. frequency, application type, and required gain.
Intelligent antenna recommendation	A recommendation algorithm (decision tree / lookup table) processes inputs and selects the most suitable antenna.
Antenna comparison	Users compare the recommended antenna against alternatives side by side.
Antenna characteristics & radiation patterns	The product displays antenna characteristics and radiation patterns for the recommendation and its alternatives.

4. Product Scope

Component	Description
Input Interface	A clean form to input technical parameters (F-001).
Core Algorithm	Embedded recommendation logic (F-002) to process inputs and suggest a primary antenna.
Visual Output	Display of the recommendation and alternatives, plus visual polar plots for the radiation pattern (F-003).

Component	Description
Comparison Feature	A side-by-side comparison table of key technical metrics (F-004).

5. Functional Requirements

ID	Priority	Requirement	Acceptance Criteria
F-001	High	Technical Parameter Input — the system must allow the user to input required technical values (frequency, application type, required gain).	All input fields support validation to ensure values are within an acceptable range (e.g. frequency ≥ 1.0 GHz).
F-002	High	Recommendation Algorithm — the system must process input data through a recommendation algorithm (decision tree / lookup table) to select the most suitable antenna.	The final recommendation must be $\geq 95\%$ accurate according to SME input and benchmarking data.
F-003	High	Visual Results Display — the system must display one primary recommendation and one alternative, with clear, interactive plots.	Plots must be clear, interactive where possible, and support image export.
F-004	Medium	Antenna Comparison — the system must let users compare the recommended antenna and two alternatives side by side in a single table.	The comparison table must include at least 5 key technical metrics (gain, beamwidth, polarization, size, cost).

6. Non-Functional Requirements

ID	Priority	Requirement	Acceptance Criteria
N-001	High	Performance — the system must complete the recommendation process and display results within a maximum of 3 seconds after the “Recommend” button is pressed.	Latency ≤ 3 seconds under normal load conditions.
N-002	High	Usability — the UI/UX must be intuitive, requiring no more than 3 steps to reach the final recommendation result.	Task completion rate $\geq 90\%$ during usability testing.
N-003	High	Compatibility — the tool must be fully compatible with all modern web browsers (Chrome, Firefox, Edge) with a responsive design for tablets and mobile.	Zero display errors when tested across screen widths.
N-004	Medium	Maintainability — the source code must be well-organized and well-commented so a future	Code comment percentage $\geq 30\%$.

ID	Priority	Requirement	Acceptance Criteria
		maintenance team can update the algorithm and database.	

7. Out of Scope / Exclusions

- Laboratory testing — no physical construction, measurement, or testing of antennas; the tool's data relies solely on published technical data and simulations.
- Real-time data updates — antenna data is static and based on currently known data; real-time updates from industry databases are excluded.
- Advanced features — user login or integration with external APIs are excluded.

8. Success Metrics

- Recommendation accuracy $\geq 95\%$ (F-002).
- System response time (latency) ≤ 3 seconds (N-001).
- Task completion rate $\geq 90\%$ during usability testing (N-002).
- Formal acceptance of the final deliverable by the Sponsor/Client.

9. Release Milestones

Milestone	Target
System design, database schema, and algorithm logic finalized	Planning (Resources) phase
Core recommendation logic implemented (Phase 1 & 2)	Execution phase
Responsive UI and comparison feature implemented	Execution phase
System integration and User Acceptance Testing (UAT)	Execution phase
Final submission and client acceptance sign-off	Closing phase

10. Dependencies

- Availability of validated antenna technical datasets for the recommendation algorithm.
- SME review and sign-off on recommendation logic accuracy.
- Timely completion of UI/UX wireframes ahead of front-end implementation.